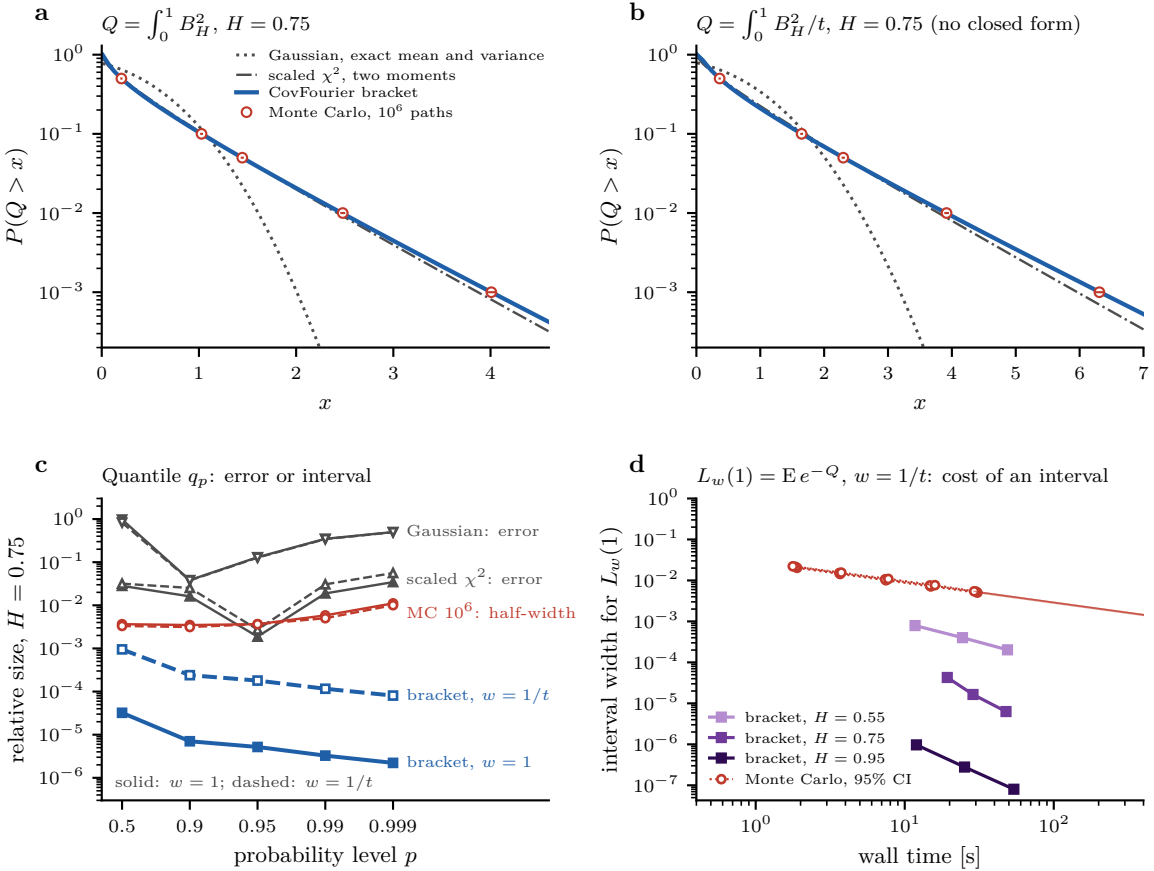

a
$Q = \int_0^1 B_H^2$, $H = 0.75$
Gaussian, exact mean and variance
scaled χ^2, two moments
CovFourier bracket
Monte Carlo, 10^6 paths
$P(Q > x)$
x
b
$Q = \int_0^1 B_H^2/t$, $H = 0.75$ (no closed form)
$P(Q > x)$
x
c
Quantile q_p: error or interval
relative size, $H = 0.75$
Gaussian: error
scaled χ^2: error
MC 10^6: half-width
bracket, $w = 1/t$
bracket, $w = 1$
solid: $w = 1$; dashed: $w = 1/t$
probability level p
d
$L_w(1) = \mathrm{E}\, e^{-Q}$, $w = 1/t$: cost of an interval
interval width for $L_w(1)$
bracket, $H = 0.55$
bracket, $H = 0.75$
bracket, $H = 0.95$
Monte Carlo, 95% CI
wall time [s]